

# Contact Book Management in C

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## 1. Abstract / Introduction

This project is a simple contact book management system developed in C. It enables users to add, view, search, edit, delete, and sort contact records. This mini-project serves as a basic command-line-based database to demonstrate structs, arrays, string handling, and basic CRUD operations.

## 2. Objective

The objective of this project is to build a basic contact book system in C which stores contact details like name, phone number, and email, and performs operations like adding, searching, updating, deleting, and sorting contacts.

## 3. Technologies Used

- Programming Language: C
- Compiler: GCC / Turbo C
- IDE: Code::Blocks or any C IDE

## 4. System Design Overview

The system follows a modular design. A `struct Contact` is defined for storing contact details. A global array holds all contacts. Functions implement actions like add, view, search, edit, delete, and sort contacts.

## 5. Features

- Add new contact
- View all contacts
- Search contact by name
- Edit phone and email of a contact
- Delete contact by name
- Sort contacts by name

## 6. Code Modules Explained

1. `addContact()`: Adds a new contact
2. `viewContacts()`: Displays all contacts
3. `searchContact()`: Finds contact by name
4. `editContact()`: Updates a contact
5. `deleteContact()`: Removes a contact from list

6. sortContacts(): Sorts alphabetically
7. main(): Menu-driven interface calling above functions

## 7. Sample Input/Output

--- Contact Book Menu ---

1. Add Contact
2. View Contacts
3. Search Contact
4. Edit Contact
5. Delete Contact
6. Sort Contacts
7. Exit

Enter your choice: 1

Enter Name: mohan rao

Enter Phone: 7019673176

Enter Email: mohanrao1622@gmail.com

Contact added successfully.

Enter your choice: 2

--- Contact List ---

Name: mohan rao

Phone: 7019673176

Email: mohanrao1622@gmail.com

## 8. Conclusion

The contact book system efficiently demonstrates structured programming in C. It showcases the usage of structures, arrays, and string functions to build a mini-application with real-world utility.

## 9. Future Enhancements

- Add file storage using file I/O to save contacts permanently
- Implement search by phone number/email
- Add GUI using C++ or integrate with a web interface
- Password protection for user privacy